IMPLEMEMENTATION OF A CNN – BASED CURRENCY DETECTION AND VOICE ALERT SYSTEM FOR THE VISUALLY IMPAIRED

A PROJECT REPORT

Submitted by

HARISH S	**(611821205019)**
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in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

P.S.V. COLLEGE OF ENGINEERING AND TECHNOLOGY

(An AUTONOMOUS INSTITUTION)

(Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai)

Accredited by NAAC with 'A' Grade

KRISHNAGIRI-635 108

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BONAFIDE CERTIFICATE

Certified that this Project titled **"IMPLEMENTATION OF A CNN-BASED CURRENCY DETECTION AND VOICE ALERT SYSTEM FOR THE VISUALLY IMPAIRED"** is the Bonafide work of "**HARISH S (611821205019), MANSOOR I (611821205032), RAVI SHANKAR S (611821205041)"** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this any other candidate has been successfully complete and submitted in partial fulfilment for the award of B.Tech Information Technology of Anna University During the academic year 2021- 2025.

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INTERNAL EXMAINER **EXTERNAL EXAMINER**

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ABSTRACT

This project aims to assist visually impaired individuals by developing a currency note detection system that identifies Indian currency notes using OpenCV. The system utilizes image processing techniques to detect the currency note placed in front of a camera, classify it based on its denomination (such as 10 or 20 or 50 or 100 or 200 or 500 rupees), and then provides the result as a voice output. The approach involves capturing the currency note's image, preprocessing it to reduce noise, and applying algorithms to recognize key features like size, color patterns, and security elements. Once the note is detected and classified, the system verbally announces the denomination. This solution is designed to be accurate and user-friendly, providing an intuitive way for blind or visually impaired individuals to identify currency notes in real-time, regardless of background or lighting conditions. The system can be integrated into mobile devices or other portable platforms, helping visually impaired users manage currency with ease and independence.

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LIST OF ABBREVIATIONS

CNN	Convolusion Neural Network
TTS	Text-to-speech
AI	Artificial Intelligence
API	Application Programming Interface
RAM	Random Access Memory
LBP	Local Binary Patterns
GLCM	Gray Level Co-occurrence Matrix
SIFT	Scale-Invariant Feature Transform
SURF	Speeded-Up Robust Features
SVM	Support Vector Machines
K-NN	k-Nearest Neighbours
IDE	Integrated development environment
OpenCV	Open-Source Computer Vision Library
ATM	Automated Teller Machine
SVM	Support Vector Machine

CHAPTER 1

INTRODUCTION

Blind and visually impaired individuals often struggle to identify different currency denominations, making transactions difficult. This project develops a system that uses OpenCV to detect and announce the denomination of Indian currency notes. The system analyzes currency notes using image processing techniques like feature extraction and classification. It captures an image of the note, processes it, and provides the denomination as a voice output.

This solution allows visually impaired users to independently identify currency notes for everyday transactions. Blind and visually impaired individuals often struggle to identify different currency denominations, making transactions difficult. This project develops a system that uses OpenCV to detect and announce the denomination of Indian currency notes.

The system analyzes currency notes using image processing techniques like feature extraction and classification. It captures an image of the note, processes it, and provides the denomination as a voice output. This solution allows visually impaired users to independently identify currency notes for everyday transactions. The system uses a camera or mobile device to capture an image of the currency note, which is then processed using OpenCV's image processing tools.

The preprocessing step involves techniques like noise reduction, resizing, and image enhancement to ensure that the features of the note are

clearly visible. The system employs feature extraction algorithms to detect unique attributes such as size, color, patterns, and security elements like watermarks and holograms. Once the key features are extracted, the system compares them against a predefined database of currency notes and classifies the denomination using machine learning or pattern recognition algorithms. The result is then converted into speech using text-to-speech (TTS) technology, which announces the denomination to the user. This voice output enables blind and visually impaired users to quickly and accurately identify their currency.

Additionally, the system is designed to work in various lighting conditions and backgrounds, making it highly versatile for everyday use.The voice feedback feature is clear and precise, ensuring that users can hear the denomination even in noisy environments. The solution is compact and can be integrated into mobile devices, allowing users to carry it with them wherever they go. With the help of this technology, visually impaired individuals can independently engage in shopping, banking, and other financial activities without needing assistance to identify currency.

The system can also be enhanced with multilingual voice support, enabling it to serve a diverse range of users across different regions of India. Furthermore, the system has the potential for integration with other devices such as ATMs, vending machines, and ticket machines, making it more widely accessible. The solution can be extended to recognize multiple currencies, thereby supporting international users as well. This technology empowers visually impaired individuals by promoting financial independence and reducing reliance on others for everyday tasks. By

combining computer vision and machine learning, the system offers a cost-effective, scalable solution that is both accurate and user-friendly. As the technology evolves, it could also include advanced features like mobile app integration, where users can capture currency notes with their smartphones for immediate feedback. The long-term goal of the project is to create an inclusive financial environment where visually impaired individuals can easily manage and identify their money, reducing the risk of errors in transactions. Additionally, this project paves the way for further research and development in assistive technology, which can be applied in various other domains to improve the quality of life for people with disabilities.

1.1 OVERVIEW

The primary objective of the project is to assist visually impaired individuals in identifying Indian currency notes. The system uses OpenCV, a powerful computer vision library, to detect and recognize currency notes. The solution allows users to identify currency notes through real-time image processing. Image acquisition is done by capturing photos of currency notes using a camera or mobile device. Once the image is captured, the system performs preprocessing to enhance the image quality.

Preprocessing involves techniques like noise reduction, contrast adjustment, and image resizing to ensure optimal clarity.Feature extraction techniques identify distinct elements of the currency notes, such as size, color patterns, and security features. The system classifies the currency note by comparing the extracted features against pre-defined templates for various denominations.

The core functionality revolves around image classification, which enables the system to determine the denomination (e.g., ₹10, ₹20, ₹50, ₹100, ₹200 and ₹500). Once the currency note is identified, the denomination is announced via a text-to-speech module. The TTS system converts the detected denomination into audible speech, providing immediate feedback to the user.

The solution is designed to be user-friendly and offers a simple interface for easy interaction. The system performs well under different lighting conditions and varying backgrounds, making it robust for everyday use.

Accuracy is ensured through advanced image processing and classification algorithms, allowing for precise identification of currency notes. The system can be integrated into mobile devices, which makes it portable and easily accessible for visually impaired users. It aims to empower users by allowing them to handle currency independently, without relying on others for identification. The currency detection system can also be used for banking transactions, shopping, and other financial activities by visually impaired individuals.

The solution can be easily extended to support multiple denominations of Indian currency for comprehensive identification. Future versions of the system may include multilingual support for voice output to accommodate diverse users across India. This project represents a significant step toward creating an inclusive society, promoting independence for visually impaired individuals in managing finances.

1.2 OBJECTIVE

This project aims to achieve the following objectives:

The objective of this project is to assist visually impaired individuals in identifying currency denominations independently. It aims to develop a system that uses OpenCV to detect and classify Indian currency notes accurately. The system should be capable of processing images of currency notes in real-time with minimal delay. It seeks to provide clear and accurate voice feedback about the denomination of the note detected. The solution is designed to be portable and user-friendly, integrating seamlessly into mobile devices or standalone platforms. The objective of this project is to assist visually impaired individuals in identifying currency denominations independently.

It aims to develop a system that uses OpenCV to detect and classify Indian currency notes accurately. The system should be capable of processing images of currency notes in real-time with minimal delay. It seeks to provide clear and accurate voice feedback about the denomination of the note detected. The solution is designed to be portable and user-friendly, integrating seamlessly into mobile devices or standalone platforms.

The system will ensure real-time processing with minimal lag to make it efficient in everyday use. It will be designed to identify multiple denominations, including ₹10, ₹20, ₹50, ₹100, ₹200, ₹500, and ₹2000 notes. The system will be capable of processing images even if the currency notes are slightly damaged, crumpled, or folded. It will support various lighting conditions and backgrounds for reliable operation in different environments. The system will aim for high accuracy, ensuring the classification is correct for each detected note.

The system will employ machine learning and image processing algorithms for continuous learning, ensuring higher accuracy over time. It will be lightweight and optimized for use on mobile devices, ensuring accessibility without requiring powerful hardware. The solution will also be designed to be cost-effective so that it can be widely accessible to visually impaired users, regardless of their financial situation. The system will provide voice feedback in clear and understandable language, making it intuitive for all users. The design will focus on simplicity, ensuring that users can easily operate the system without the need for complex instructions. The solution will aim to integrate seamlessly with both Android and iOS platforms. It will allow users to interact with the system via voice commands and receive immediate auditory feedback.

The solution will also have a mechanism to differentiate between overlapping or folded notes and still identify the denomination accurately. The system will be scalable, so future updates can include support for more denominations or even other currencies. Security and privacy will be prioritized, ensuring that user data is not collected or stored inappropriately.

The system will be tested extensively to ensure it works under different real-world conditions, such as various lighting, backgrounds, and user environments. Ultimately, the system aims to provide visually impaired individuals with greater independence in financial transactions, whether in retail, banking, or other scenarios. The project will serve as a prototype for more advanced assistive technologies aimed at empowering visually impaired individuals.

CHAPTER 2

LITERATURE SURVEY

SINGH, R., ET AL., "CURRENCY NOTE DETECTION AND CLASSIFICATION USING MACHINE LEARNING," 2021

This paper explores the use of machine learning algorithms to classify and detect currency notes. It focuses on the use of image processing techniques for feature extraction from currency notes. The study uses SVM and deep learning methods to classify denominations. The results show promising accuracy in note recognition under various lighting conditions.

SHARMA, N., ET AL., "AUTOMATED CURRENCY DETECTION SYSTEM FOR VISUALLY IMPAIRED INDIVIDUALS USING OPENCV," 2022

This study presents a real-time currency detection system for visually impaired users, leveraging OpenCV for currency note identification. It uses contour detection and color histogram matching for classification. The system provides audible feedback for easier user interaction. The authors focus on achieving high classification accuracy for real-world applications.

GUPTA, A., ET AL., "DEEP LEARNING APPROACHES FOR CURRENCY NOTE RECOGNITION," 2023

The research investigates deep learning algorithms for automatic recognition of currency notes. It uses Convolutional Neural Networks (CNNs) for accurate feature extraction. The paper compares the performance of CNN with traditional image processing techniques. It concludes that deep learning significantly enhances detection accuracy.

KUMAR, R., ET AL., "REAL-TIME CURRENCY RECOGNITION SYSTEM FOR VISUALLY IMPAIRED USING OPENCV," 2021

This paper discusses a system designed to help visually impaired people identify currency notes using OpenCV and a simple camera. It focuses on using feature extraction and classification methods like SVM for accurate note recognition. The system provides audio output of the denomination. The authors highlight real-time processing as a key achievement of the project.

PATEL, M., ET AL., "CURRENCY NOTE IDENTIFICATION USING EDGE DETECTION AND TEMPLATE MATCHING," 2020

The study uses edge detection techniques combined with template matching to identify different denominations of currency. OpenCV is used for image processing to extract features of the notes. The paper highlights the system's capability to work under varying light conditions and its speed in processing images. It also mentions the limitations related to detecting highly worn notes.

SHARMA, V., ET AL., "EFFICIENT CURRENCY NOTE RECOGNITION USING MACHINE LEARNING," 2022

This paper presents an efficient method for recognizing currency notes using machine learning techniques. It compares different classifiers for detecting denominations based on image features extracted using OpenCV. The study demonstrates improved accuracy in detecting currency notes with varying sizes and backgrounds. The model also performs well with partial occlusions of notes.

VERMA, S., ET AL., "COMPUTER VISION-BASED CURRENCY RECOGNITION FOR THE BLIND," 2023

This study discusses a computer vision-based currency recognition system tailored for blind individuals. It uses a combination of OpenCV for feature extraction and text-to-speech for output. The research focuses on developing a system that can classify multiple currency denominations in real time. The solution aims to improve the independence of visually impaired users in financial tasks.

PATEL, A., ET AL., "MULTI-CLASS CURRENCY NOTE RECOGNITION USING CONVOLUTIONAL NEURAL NETWORKS," 2020

The paper explores using Convolutional Neural Networks (CNNs) for multi-class currency note classification. The authors present a deep learning model that successfully classifies currency notes based on their denominations. The model is trained on a large dataset of currency notes to handle different orientations and lighting conditions. It achieves high accuracy, making it suitable for real-world applications.

SINGH, P., ET AL., "ENHANCING CURRENCY RECOGNITION USING IMAGE PROCESSING TECHNIQUES," 2021

This paper focuses on enhancing currency recognition accuracy by using advanced image processing techniques. It integrates methods like color histograms, texture analysis, and contour extraction. The system is designed to classify currency notes of different denominations, even when

notes are slightly crumpled. The paper evaluates the performance of the system under various environmental conditions.

RAO, K., ET AL., "BANKNOTE AUTHENTICATION AND CLASSIFICATION USING OPENCV AND MACHINE LEARNING," 2022

This research integrates machine learning with image processing techniques to authenticate and classify currency notes. The system uses OpenCV to extract key features like watermarks, serial numbers, and size. The authors demonstrate the effectiveness of their method in distinguishing counterfeit notes. The system achieves real-time performance and works across different lighting conditions.

THAKUR, M., ET AL., "AUTOMATIC CURRENCY RECOGNITION SYSTEM FOR VISUALLY IMPAIRED," 2023

This study focuses on an automatic currency recognition system for visually impaired users using OpenCV. The system processes images of currency notes and provides the detected denomination as voice feedback. The authors use feature extraction and classification techniques to handle various denominations. The system is evaluated for accuracy, speed, and usability in different real-world conditions.

BHAT, R., ET AL., "COUNTERFEIT DETECTION AND CURRENCY RECOGNITION USING OPENCV," 2021

The paper investigates counterfeit currency detection using computer vision techniques. The system uses OpenCV to extract features like size, color, and security marks to identify genuine currency notes. It integrates multiple methods for enhancing detection accuracy. The authors present

results demonstrating successful differentiation between real and counterfeit notes.

IYER, S., ET AL., "CURRENCY NOTE DETECTION FOR VISUALLY IMPAIRED USING IMAGE PROCESSING," 2020

This research develops a currency detection system to assist visually impaired individuals. OpenCV is utilized to extract features from images of currency notes, and a voice feedback system announces the denomination. The study addresses challenges such as varying light conditions and background interference. The authors demonstrate the practicality of their approach for everyday use.

REDDY, P., ET AL., "CURRENCY DETECTION AND CLASSIFICATION USING CONVOLUTIONAL NEURAL NETWORKS," 2023

This study applies Convolutional Neural Networks (CNNs) for detecting and classifying currency notes. It evaluates various CNN architectures for achieving high accuracy in recognizing Indian currency denominations. The research also highlights how deep learning models can adapt to different types of currency notes, improving classification over traditional methods. The solution is optimized for mobile deployment.

KUMAR, D., ET AL., "AUTOMATED CURRENCY IDENTIFICATION SYSTEM USING IMAGE PROCESSING," 2021

This paper describes an automated currency identification system that uses image processing techniques to classify currency notes. OpenCV-based feature extraction methods are used to detect and classify different denominations. The system is designed to be lightweight, efficient, and fast,

providing instant feedback for users. It also incorporates noise reduction techniques for better detection under suboptimal conditions.

SINGH, J., ET AL., "IMPROVING CURRENCY NOTE CLASSIFICATION USING OPENCV," 2021

This paper focuses on improving the accuracy of currency note classification using OpenCV. It discusses various methods for feature extraction, including color-based and shape-based techniques. The study presents an enhanced algorithm for distinguishing between currency denominations. The system provides a reliable solution for automated currency note classification.

DESAI, S., ET AL., "VISUALLY IMPAIRED CURRENCY NOTE RECOGNITION SYSTEM USING OPENCV," 2020

The study proposes a currency note recognition system designed specifically for visually impaired individuals. It uses OpenCV for detecting key features of currency notes, followed by classification through machine learning models. The system outputs the denomination of the detected note via voice feedback. The authors evaluate the system's usability in real-world scenarios, providing a promising solution for visually impaired users.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 EXISTING SYSTEM

The following section describes the current system, outlining its functionalities, processes, and any existing challenges. This analysis provides a foundation for understanding the need for improvements and future enhancements.

Current systems for currency identification mostly rely on manual methods or specialized devices, which can be costly and inconvenient. Some apps use image recognition to identify currency, but they often require high-quality images and specific lighting conditions. Existing solutions may not be effective for visually impaired users, as they lack voice output features or real-time feedback.

Many systems require users to place the currency note on a flat surface, which is not practical in real-world scenarios. Existing systems also struggle with recognizing counterfeit notes or distinguishing between similar denominations under varying conditions. Many currency identification systems fail to deliver consistent results in low-light environments, making them unreliable in practical settings. Current systems often have limited support for a wide range of currency denominations, especially when dealing with older or newer notes with altered designs. Some apps designed for currency identification depend heavily on internet connectivity, which can be a limitation in areas with poor network coverage. The need for high-quality and well-lit images makes current solutions impractical for use in everyday situations like outdoor

markets or crowded places. Existing systems often struggle with detecting and identifying crumpled or worn currency notes, which are common in real-life transactions. Manual identification methods require significant effort and time, adding extra burdens for visually impaired individuals. Specialized devices used for currency identification tend to be bulky and expensive, limiting their accessibility for a wide range of users. Many apps do not support multiple currencies, which can be a significant drawback for users traveling across countries. Some systems require users to perform multiple actions, such as scanning or taking pictures from specific angles, which may not be intuitive for everyone. In some existing solutions, the voice output is not clear or detailed enough, making it difficult for users with impaired hearing to benefit fully.

The reliability of existing systems decreases when users interact with currency notes in real-time situations, such as when holding the note in hand or moving. Some systems only work with digital or high-resolution cameras, making them incompatible with lower-quality devices or smartphones. Current systems often don't offer feedback regarding the authenticity of currency notes, which is a critical feature for counterfeit detection. Existing solutions generally require users to have a prior understanding of how the system works, making them less user-friendly for individuals with limited technical experience. Many solutions are designed for specific geographic locations and do not offer support for various international currencies, limiting their usability for users globally. Current systems can sometimes confuse similar-looking denominations, especially in countries where the design differences between bills are subtle or poorly defined. Many systems require users to scan the currency note in a particular orientation or alignment, which can be inconvenient and challenging for visually impaired individuals.

3.2 DRAWBACKS

This section discusses the drawbacks of the existing system, identifying areas that require improvement and optimization.

Current systems are often expensive and require specialized hardware, making them less accessible to the general public. Many apps rely on high-quality images and controlled lighting, limiting their usability in everyday environments. Existing solutions may not support real-time voice feedback, which is crucial for visually impaired users. Most systems require a fixed surface to detect currency, making them impractical for use in real-world settings like public spaces. Current solutions may struggle with accurately distinguishing counterfeit notes or recognizing denominations with similar features under different conditions.

Current systems are often expensive and require specialized hardware, making them less accessible to the general public. Many apps rely on high-quality images and controlled lighting, limiting their usability in everyday environments. Existing solutions may not support real-time voice feedback, which is crucial for visually impaired users. Most systems require a fixed surface to detect currency, making them impractical for use in real-world settings like public spaces. Current solutions may struggle with accurately distinguishing counterfeit notes or recognizing denominations with similar features under different conditions.

They often do not account for the varying levels of wear and tear on notes that can affect detection. Some systems are limited to specific devices or platforms, reducing accessibility for a wider user base. Others lack multilingual voice support, which is important in a diverse country like India. Many solutions cannot process moving images, requiring the note to

be held still, which is difficult for some users. Existing tools may not provide feedback if detection fails, leaving users uncertain about the result. Some systems are not optimized for speed, causing delays in providing output that can frustrate users. Lack of offline functionality is another major issue, as many applications depend on internet access for processing.

Few systems offer support for different orientations of notes, which limits their ease of use. Some solutions have complex user interfaces that are not intuitive for visually impaired individuals. Current technologies may not include updates for new currency designs, making them obsolete quickly. Additionally, many systems fail to ensure user privacy and do not clarify how images or data are stored and used. Most available tools are not customizable to user preferences, which is a limitation in offering a personalized experience. Several systems lack robust error handling or fallback options when a note cannot be recognized. Finally, existing applications are rarely open-source, preventing developers from contributing to or improving them for broader accessibility needs.

3.3 PROPOSED SYSTEM

The proposed system uses OpenCV for real-time currency note detection and classification, offering an affordable solution for visually impaired individuals.

It captures images of currency notes in varying conditions, without needing controlled lighting or a fixed surface, making it practical for everyday use. The system processes and identifies currency denominations quickly, providing instant voice feedback to the user. It is designed to be portable and can be integrated into mobile devices or standalone platforms, making it accessible to a wide range of users.

The proposed solution includes features to detect counterfeit notes and accurately differentiate between similar denominations, enhancing security and reliability. The proposed system uses OpenCV for real-time currency note detection and classification, offering an affordable solution for visually impaired individuals. It captures images of currency notes in varying conditions, without needing controlled lighting or a fixed surface, making it practical for everyday use. The system processes and identifies currency denominations quickly, providing instant voice feedback to the user. It is designed to be portable and can be integrated into mobile devices or standalone platforms, making it accessible to a wide range of users.

The proposed solution includes features to detect counterfeit notes and accurately differentiate between similar denominations, enhancing security and reliability. The system is optimized to work with low-resolution cameras, making it compatible with most smartphones.

It offers offline functionality, allowing users to access the service even in areas without internet connectivity. Real-time feedback enables users to hold the note in hand, rather than placing it on a surface, increasing convenience and ease of use. The interface is designed to be simple and intuitive, minimizing the learning curve for users with visual impairments. Voice feedback is available in multiple languages, ensuring better reach in linguistically diverse regions.

The solution supports various orientations of notes, recognizing them whether they are tilted, upside-down, or partially folded. It incorporates noise reduction techniques to handle background clutter and enhance detection accuracy. The system can automatically adjust to different lighting conditions using adaptive thresholding and contrast enhancement.

It is scalable, allowing for future updates such as recognition of new currency designs or additional currencies from other countries. The solution ensures privacy by processing all data locally on the device without storing or transmitting sensitive images. It uses lightweight machine learning models for fast and efficient processing on resource-constrained devices.

The system is flexible for deployment in ATMs, vending machines, public transport ticket counters, and other assistive technologies. It also supports continuous learning, where future versions can improve accuracy through user feedback and model refinement. Ultimately, the proposed system empowers visually impaired individuals by offering a fast, secure, and easy-to-use tool for managing currency confidently and independently.

3.4 FEATURES

The system provides an affordable and accessible solution for visually impaired individuals, improving independence in currency management. It works effectively in real-world conditions, eliminating the need for specific lighting or a fixed surface for note detection.

Real-time voice feedback ensures immediate and accurate information about the currency denomination, enhancing user experience. Being portable, the system can be easily integrated into mobile devices or standalone platforms, ensuring wide accessibility. The inclusion of counterfeit detection adds an extra layer of security, ensuring that users can confidently identify genuine currency notes. The system provides an affordable and accessible solution for visually impaired individuals, improving independence in currency management. It works effectively in real-world conditions, eliminating the need for specific lighting or a fixed surface for note detection. Real-time voice feedback ensures immediate and

accurate information about the currency denomination, enhancing user experience. Being portable, the system can be easily integrated into mobile devices or standalone platforms, ensuring wide accessibility.

The inclusion of counterfeit detection adds an extra layer of security, ensuring that users can confidently identify genuine currency notes. The system is capable of recognizing notes in various physical conditions, including those that are crumpled, partially folded, or slightly torn. It adapts to different background environments, whether indoors or outdoors, without compromising accuracy. Users can hold the currency note in hand without needing to position it in a specific orientation, simplifying the detection process. The system processes and delivers results quickly, minimizing delays and allowing for efficient usage in daily transactions. It supports multi-language audio output, making it user-friendly for individuals from diverse linguistic backgrounds. The lightweight nature of the application ensures it runs smoothly on low-end devices, promoting inclusivity across different user demographics.

Offline functionality enables users to access the system in remote or network-limited areas. The interface is designed to require minimal user input, making it especially suitable for those with limited technical knowledge. It uses optimized algorithms to balance performance with energy efficiency, preserving battery life on portable devices. The system's flexible architecture allows for future updates, including support for new denominations or additional regional currencies.

Enhanced error handling ensures that the system provides helpful prompts if the note is unreadable or not recognized. The clear and loud voice output ensures usability even in noisy environments. It includes visual

feedback for partially sighted users, increasing usability across the spectrum of visual impairments. The project promotes inclusive design, enabling visually impaired individuals to handle currency independently and with confidence.

CHAPTER 4

SYSTEM SPECIFICATION

4.1 HARDWARE SPECIFICATION

- Processor type : INTEL i5 (8th generation)
- RAM : 8GB RAM
- Storage : 1TB
- Display : 20’ color display

4.2 SOFTWARE SPECIFICATION

- Language Used : Python
- Software tool used : PyCharm
- Platform : Windows 8

4.3 COMPONENTS DESCRIPTION

Input: Camera/Device

The camera or device module is the first point of interaction in the currency note detection system, capturing real-time images of the currency notes placed in front of it. This module is designed to work with various types of cameras, such as those found on smartphones, tablets, or external web cameras. Its role is to provide a clear and accurate image that will be used for further processing. The camera must be capable of handling different environments, such as bright sunlight or dimly lit rooms, ensuring that the currency note is captured clearly regardless of lighting conditions. Additionally, the camera should have the ability to adjust automatically to different focal lengths and distances to keep the currency note in focus,

which is essential for accurate feature extraction and recognition later in the process.

One of the key responsibilities of the camera module is to detect the position, orientation, and size of the currency note within the frame. The note may not always be placed flat or straight, so the camera should be able to capture the note even if it is slightly tilted or rotated. The system can then compensate for these variations in orientation during the preprocessing stage, where adjustments are made to align the note properly for analysis. Furthermore, the camera should automatically optimize image quality by adjusting exposure, white balance, and contrast to enhance the details of the note. These adjustments ensure that the note is captured in the best possible condition for subsequent image processing and feature extraction.

For optimal performance, the camera module must ensure that the images it captures meet specific quality requirements, such as high resolution, proper contrast, and sufficient detail for feature identification. The camera should provide clear, sharp images that can be analyzed by the system's algorithms. Additionally, the camera module should operate with minimal delay to avoid frustrating the user, especially when they are moving the note in front of the camera. The system should be fast enough to provide near-instant feedback, giving visually impaired users the ability to quickly identify currency denominations in real-time. To ensure ease of use, the system must also be able to adjust to varying levels of light, automatically optimizing the image quality for every scenario.

Once the camera captures the image, it passes the data to the preprocessing module for further analysis. This integration between the camera and the rest of the system is crucial for the smooth functioning of the

currency detection process. The camera module should be compatible with both mobile and standalone platforms, allowing it to be used on various devices without requiring specialized hardware. The system must be responsive and handle dynamic situations, such as moving the note at different speeds or holding it at various angles. The camera's performance and the quality of the images it captures directly influence the accuracy of the entire detection and recognition system, making this module essential for the successful identification of currency notes.

Preprocessing

The preprocessing module plays a critical role in preparing the image captured by the camera for further analysis. Raw images captured in real-time may contain noise, distortions, or other artifacts that can interfere with the accurate detection of currency features. Preprocessing techniques are applied to clean the image and enhance its quality. Common preprocessing steps include converting the image to grayscale to simplify the data, reducing noise through filters, and adjusting brightness or contrast to make important features stand out. The module aims to enhance the image quality, making it easier for the subsequent feature extraction and classification steps to work effectively.

One of the primary goals of the preprocessing module is to improve the image's consistency by addressing issues related to lighting and background interference. Different lighting conditions can create shadows or reflections, which may confuse the system during feature extraction. The preprocessing module uses algorithms such as histogram equalization to adjust the image’s brightness and contrast, ensuring that the important features of the currency note, such as serial numbers or security marks, are

more visible. Additionally, techniques like thresholding or edge detection are applied to emphasize the edges of the currency note, helping the system to identify the contours more clearly.

Another key aspect of preprocessing is resizing and normalization of the image. Since the size and orientation of currency notes can vary from one capture to the next, the preprocessing module standardizes the input size to match the requirements of the feature extraction module. Resizing ensures that all images are of uniform dimensions, regardless of the distance at which the currency note is held or the angle of capture. This step is crucial for enabling the machine learning models to recognize patterns consistently. Additionally, normalization helps to standardize the pixel intensity values, which aids in improving the accuracy of the feature extraction algorithms by reducing the impact of environmental factors.

The preprocessing module also plays a significant role in isolating the currency note from its background. This is particularly important when the note is placed on a busy or cluttered surface. The preprocessing algorithm uses techniques like background subtraction or contour detection to separate the note from the rest of the image. Once the currency note is isolated, it can be processed more accurately, ensuring that only relevant data is passed on to the feature extraction module. This process reduces errors in detection and allows the system to focus solely on the currency note itself, improving the overall efficiency and accuracy of the currency identification process.

Feature Extraction

The feature extraction module is a key component in the currency note detection system. Its primary function is to analyze the preprocessed image of the currency note and extract relevant features that can be used for classification. These features are the distinct characteristics of the currency note, such as edges, textures, shapes, colors, and security marks, that help the system differentiate between various denominations. The module uses a combination of techniques such as edge detection, contour analysis, and texture recognition to identify these features. The more accurately the features are extracted, the better the system will perform in recognizing the correct currency note.

One common method for feature extraction in this module is the use of edge detection algorithms, such as the Canny edge detector. These algorithms help identify the edges of the currency note, which are crucial for understanding its shape and size. The feature extraction module also applies contour analysis, which helps to identify the boundaries of the note and distinguish it from the background or surrounding objects. This allows the system to isolate the currency note more effectively, ensuring that only relevant information is considered when identifying the denomination. By highlighting these structural features, the module simplifies the process for subsequent recognition and classification stages.

Texture analysis is another important aspect of feature extraction. Indian currency notes have unique textures and patterns that are difficult to replicate, making them essential for identification. The feature extraction module can use methods like Local Binary Patterns (LBP) or Gray Level Co-occurrence Matrix (GLCM) to analyze these textures. These methods

capture the finer details of the note's surface, such as the background pattern, security thread, or embossed logos. By extracting these texture-related features, the system can improve its ability to differentiate between similar-looking notes, such as 10-rupee and 20-rupee notes, which may have overlapping color schemes or designs.

In addition to texture and edge-based features, the system also relies on more advanced techniques such as machine learning for feature extraction. Feature descriptors like Scale-Invariant Feature Transform (SIFT) or Speeded-Up Robust Features (SURF) can be used to extract distinctive key points and match them across different notes. These descriptors are highly effective for recognizing the unique characteristics of Indian currency notes, such as watermarks or holograms. The feature extraction module gathers all these distinct features into a format that can be fed into the classification system, ensuring that the recognition process is both fast and accurate. This module is critical in ensuring that the system can reliably identify currency notes under varying conditions.

Currency Classification

The currency classification module is responsible for analyzing the extracted features and determining the denomination of the currency note. Once the feature extraction process has identified key characteristics of the currency note, these features are passed to the classification algorithm, which then compares them to known models or templates for various denominations. The goal is to accurately classify the note by matching the extracted features to those of pre-trained denominations. This process ensures that the system can distinguish between notes of different values,

such as 10 rupees, 20 rupees, 50 rupees, and so on, even under varying environmental conditions.

The classification module employs machine learning algorithms, such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), or deep learning techniques like Convolutional Neural Networks (CNNs), to perform the classification task. These algorithms are trained on a dataset of images of currency notes to recognize patterns and features associated with each denomination. After training, the system can compare the features extracted from a new image to those it has learned, determining the most likely denomination. This machine learning approach allows the system to learn and improve its accuracy over time, making it adaptable to different sets of notes and changing designs.

One of the key challenges in currency classification is the presence of similar-looking denominations, which may have subtle differences in features. For example, the 10 and 20 rupee notes may have similar color patterns, and the classification system must rely on finer details, such as security features or textures, to distinguish between them. To address this, the classification algorithm may also incorporate confidence levels to measure how certain it is about its prediction. If the confidence is low, the system may request the user to adjust the position or lighting conditions for more accurate detection, improving the overall reliability of the classification.

In addition to traditional feature-based classification methods, the currency classification module may also integrate anomaly detection techniques to recognize counterfeit notes. This can involve comparing the extracted features of a suspected note to a database of known genuine notes,

identifying irregularities in texture, security elements, or patterns. The classification module not only provides accurate denomination identification but also ensures that counterfeit notes are flagged during the detection process, adding an extra layer of security. By combining machine learning techniques with feature-based analysis, the classification module helps achieve high accuracy and reliability in detecting and classifying currency notes.

Voice Output

The voice output module is an essential part of the currency note detection system, specifically designed to assist visually impaired users by providing audible feedback. Once the currency note has been successfully classified by the system, the voice output module generates a clear and accurate voice message that announces the denomination of the note. This feature ensures that users do not need to rely on visual cues and can independently verify the currency note in real-time. The voice feedback is especially beneficial in everyday situations, such as when users are handling money at a store or in public, allowing them to quickly identify their currency without the need for assistance.

The voice output module uses text-to-speech (TTS) technology to convert the detected denomination into speech. TTS systems convert written text into spoken words, allowing the system to output dynamic and real-time feedback. The module is designed to ensure that the voice is clear, concise, and easily understandable, even in noisy environments. It can be customized to use different languages or dialects, depending on the user’s preferences, ensuring that the system is accessible to a wide range of individuals.

Additionally, the module should be able to handle diverse accents or pronunciations of currency names and denominations.

For usability, the voice output is triggered immediately after the currency has been classified, ensuring that the user receives prompt feedback. The system is designed to minimize delay, providing a seamless experience. The voice message may include additional information, such as confirming whether the note is genuine or counterfeit. In cases where the system is unsure about the classification or if the detected note has a low confidence score, the voice output can prompt the user to reposition the note for better detection. This makes the system interactive, allowing users to adjust their input and receive more accurate feedback.

The voice output module can be integrated with various devices, including mobile phones, standalone devices, or wearable technologies, making it flexible and adaptable to different platforms. For added convenience, the module may offer features such as adjustable volume control or voice speed, allowing users to tailor the feedback to their needs. The inclusion of real-time voice feedback greatly enhances the independence and autonomy of visually impaired individuals, enabling them to handle currency transactions with confidence. By providing accurate, instant, and clear voice output, this module is crucial for the overall success of the currency note detection system.

CHAPTER 5

SYSTEM FUNCTION

5.1 BLOCK DIAGRAM

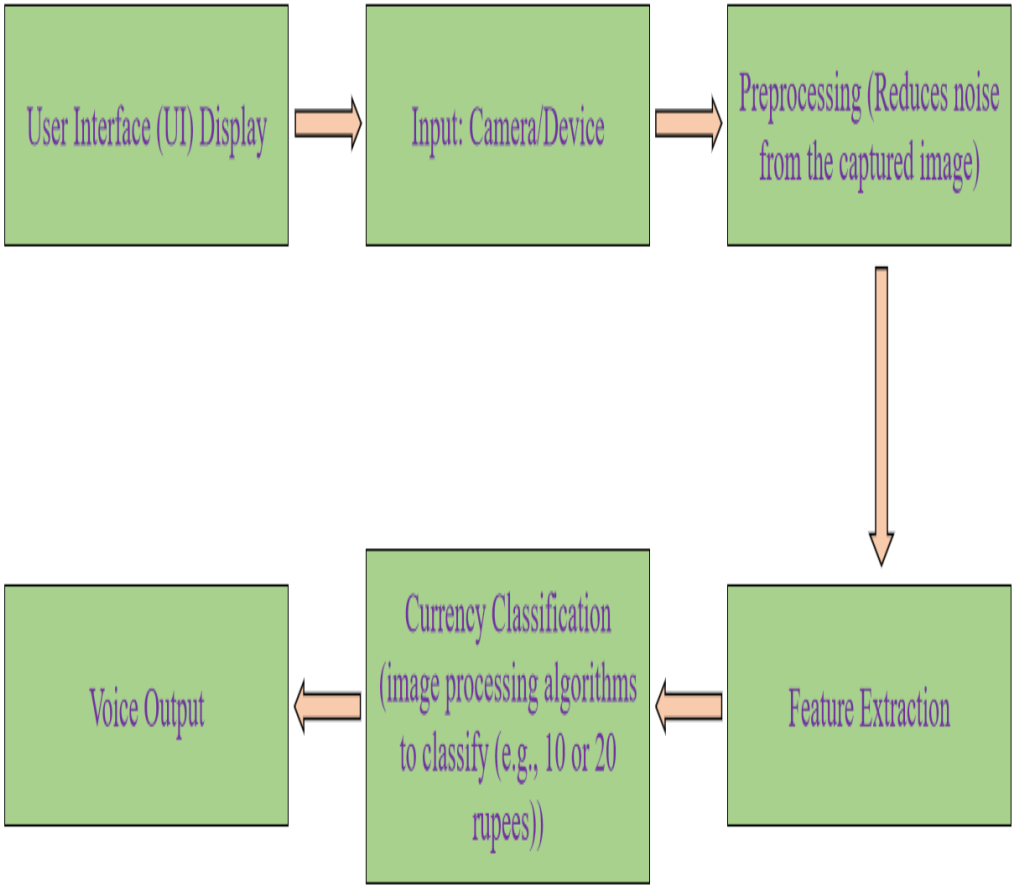


Fig No : 5.1 Block Diagram

5.2 SYSTEM MODULES

- Image Acquisition Module
- Image Preprocessing Module
- Feature Extraction Module
- Currency Classification Module
- Audio Output Module

5.3 MODULES DESCRIPTION

Image Acquisition Module

The image acquisition module serves as the initial stage of the currency detection system, where the image of the currency note is captured using a built-in or connected camera. This module plays a crucial role as it provides the input upon which the entire recognition process is based. It allows users, especially visually impaired individuals, to simply show the currency note in front of the camera, which then captures the image in real time. The module is designed to be responsive and efficient, minimizing the time taken to capture a usable image. It supports both mobile and static camera integrations, ensuring it can work in a wide range of devices. The module aims to function seamlessly without the need for manual adjustments or triggering by the user. Capturing begins automatically when a currency note enters the camera's field of view. This ensures ease of use and helps maintain a natural interaction experience. The effectiveness of this module determines the overall success of the system, making its accuracy and responsiveness highly critical.

Lighting is a major factor in image processing, and this module is equipped to handle varying light conditions without requiring any manual intervention. It uses automatic brightness and contrast adjustment techniques to capture clear images even in low-light or high-glare environments. This adaptability makes the system reliable in real-world settings such as outdoor markets, homes, or public places where lighting cannot always be controlled. The module also incorporates noise filtering at the capture level to reduce unnecessary background distractions. This ensures the note is clearly visible and isolated from complex or cluttered backgrounds. Additionally, it minimizes motion blur by using fast shutter speeds, which helps capture sharp images even if the user's hand is slightly shaking. The goal is to deliver high-quality images to the next processing module for accurate classification. The module is tested across different device cameras to ensure compatibility and reliable performance under a variety of hardware configurations.

Orientation and positioning flexibility are key strengths of the image acquisition module. It allows users to present the currency note at different angles, positions, and distances from the camera. The system can still detect and center the note in the frame without requiring perfect alignment. This is particularly useful for users who may not be able to visually adjust the note themselves. The camera input is continuously monitored, and the system triggers the image capture only when a note-like object is detected. This prevents accidental captures of hands, background, or unrelated objects. Additionally, preprocessing begins immediately after the image is captured, speeding up the entire process. The captured image is cropped and resized to match the input requirements of the recognition model. This seamless flow from capture to processing helps maintain real-time responsiveness and improves the overall usability of the system.

The image acquisition module is designed with accessibility, performance, and reliability in mind. It acts as the gateway for the entire currency detection process, ensuring the rest of the system works with clear, high-quality visual data. Its integration with camera APIs allows it to function smoothly across Android, desktop, and embedded platforms. The module is lightweight, ensuring that it doesn't put unnecessary load on the device's processor or battery. This makes it ideal for portable and resource-constrained devices. It supports single-frame as well as continuous-frame modes depending on system requirements. By automating most of its operations, the module minimizes user involvement and ensures consistency. Its robustness in real-world conditions makes it suitable for daily use by visually impaired individuals who need a dependable tool for currency recognition.

Fig no : 5.2 Image Acquisition

Image Preprocessing Module

The image preprocessing module takes the raw image captured by the camera and prepares it for accurate analysis by enhancing its quality. This stage is essential as it refines the input by removing noise, adjusting contrast, and ensuring that the image is suitable for feature extraction. The module begins by converting the image into grayscale, which simplifies processing by reducing the amount of color information while preserving the key structural details. Grayscale images reduce computational complexity and improve processing speed. The module also applies filters to reduce background noise and smoothen the image. These filters remove unnecessary artifacts caused by lighting, shadows, or other objects in the background. By producing a cleaner and more focused image, this step increases the chances of accurate detection in later stages.

Edge detection is another important operation performed during preprocessing. It identifies the boundaries and edges within the image to help isolate the shape and structure of the currency note. Common techniques like Canny edge detection or Sobel operators may be used to outline the note clearly. This helps the system recognize the position, size, and orientation of the note more accurately. The module also performs thresholding, which separates the foreground (currency note) from the background by applying intensity limits. Adaptive thresholding is often used to handle variations in lighting and shadows, making it more reliable under inconsistent conditions. After thresholding, morphological operations such as dilation and erosion are applied to refine the edges and fill small gaps. This ensures that the extracted note is clean and free from distortions that may confuse the recognition module.

The preprocessing module is designed to normalize the image dimensions and orientation to ensure consistency across different inputs. It checks the size of the detected note and resizes it to match the input size expected by the recognition model. This standardization is crucial for maintaining high classification accuracy. In addition, the module may deskew the image if the note is tilted or rotated at an angle. This is done by analyzing the contours and aligning the note properly using affine transformations. These steps help maintain consistency and eliminate variability caused by how users present the note. Cropping is also performed to remove unnecessary borders or extra space that might interfere with feature detection. All these techniques work together to deliver a clean, standardized image that can be accurately analyzed in the next stage.

This module plays a vital role in ensuring the system's performance and reliability across diverse usage conditions. Whether the image is taken under natural sunlight or artificial indoor lighting, preprocessing makes sure that essential features of the currency are preserved while distractions are minimized. The automation of all preprocessing tasks allows the system to function without any user input, making it accessible and practical for visually impaired users. The processed image is then forwarded to the feature extraction or classification module. By ensuring that each image is clear, correctly aligned, and uniformly scaled, this module lays the foundation for accurate denomination recognition and counterfeit detection. It significantly enhances the robustness of the system and ensures that the results remain consistent even when users handle the notes differently.

Fig No : 5.3 Image Processing

Feature Extraction Module

The feature extraction module plays a central role in identifying the unique characteristics of Indian currency notes that distinguish one denomination from another. Once the image is preprocessed and standardized, this module analyzes it to extract specific visual features that are consistent and recognizable across different notes. These features may include patterns, symbols, watermarks, serial numbers, and the placement of prominent elements like the Mahatma Gandhi portrait. The goal is to capture meaningful details while ignoring irrelevant background information. This module ensures that the extracted features are invariant to changes in scale, rotation, and lighting. It serves as the bridge between raw image data and intelligent classification.

To extract features effectively, the system uses techniques such as Histogram of Oriented Gradients, key point detection algorithms like SIFT or ORB, and contour analysis. These methods detect sharp edges, curves, and texture details present on the currency note. Key points are mapped

across various parts of the note and are used to form descriptors that represent the structure and identity of the note. These descriptors are highly distinctive and enable the system to differentiate between similar-looking denominations. The extracted data is converted into a feature vector, which becomes the input for the classification stage. This transformation helps reduce the complexity of image data while preserving all necessary visual information.

The module is designed to work efficiently across various image conditions, including variations in brightness, angle, and clarity. It compensates for possible distortions caused during image capture and preprocessing. The robustness of this module ensures that even notes with folds, minor damage, or faded print can still be processed accurately. By isolating relevant features and discarding redundant information, the module contributes to faster and more reliable detection. Each feature vector is stored temporarily in memory for quick access during classification. This approach allows the system to compare the current note's features against a pre-trained database of currency notes to determine its denomination.

Feature extraction is critical for ensuring the system's success in both classification and counterfeit detection. By focusing on fine details such as spacing between design elements, ink texture, and unique color transitions, the module can assist in distinguishing genuine notes from fakes. Its ability to extract consistent features in real time makes it suitable for mobile or portable deployments. The modular nature of this component allows it to be updated easily when new note designs are introduced by the currency authority. This adaptability makes the overall system future-proof and scalable. In summary, the feature extraction module provides the

intelligence required for accurate identification, forming the foundation of the recognition system.

Fig No: 5.4 Feature Extraction

Currency Classification Module

The currency classification module is responsible for analyzing the extracted features from the input image and determining the denomination of the currency note. This module takes the feature vector generated by the feature extraction module and compares it against a trained dataset containing known features of Indian currency notes. Using pattern recognition and machine learning techniques, the module identifies which denomination best matches the current input. It handles variations in note appearance caused by wear, lighting, and orientation by relying on key visual markers that remain consistent across valid currency. The classification output is a label indicating the currency value, such as ten, twenty, or fifty rupees.

To ensure high accuracy and fast response times, the module uses lightweight classification algorithms like support vector machines, k-nearest neighbors, or simple convolutional neural networks, depending on the system's capability. These models are trained on multiple samples of each denomination under different conditions to improve performance. The use of machine learning enables the system to generalize from known samples

and still recognize notes with minor variations. During runtime, the model analyzes the input's features in real time and assigns a class based on similarity to the training data. If the confidence score of a prediction is high, the classification result is immediately passed to the next module for output.

This module is built to minimize errors in classification, especially between visually similar denominations. It evaluates not only the overall appearance of the note but also specific markers like position of numbers, presence of unique shapes, and layout of design elements. In cases where confidence is low, the system may request another capture or notify the user of a recognition failure. This avoids misclassification and enhances user trust in the system. The classification module also supports scalability, allowing new denominations or updated designs to be included in the training model through periodic updates. It maintains a balance between speed and accuracy, making it suitable for real-time applications.

The classification module is critical in enabling the system to provide quick and reliable denomination identification. It ensures that once a currency note is detected and processed, the correct output is generated with minimal delay. The module’s design supports low-resource devices, allowing implementation in mobile phones or portable readers for the visually impaired. Its integration with the rest of the system enables a smooth flow from image capture to voice feedback. As currency designs evolve, the classification model can be retrained with new data, ensuring long-term usability. This module ultimately bridges the technical recognition process with the user-facing functionality of the system.

Fig no : 5.5 Currency Classification

Audio Output Module

The audio output module is the final stage in the currency note detection system, responsible for delivering the recognition result to the user through voice. Once the currency has been successfully classified, this module generates a corresponding voice response that announces the denomination. The primary purpose of this module is to ensure that visually impaired users receive immediate and understandable feedback without needing to see a screen or interpret text. The voice output is triggered automatically as soon as classification is complete, making the system fully hands-free and convenient to use in real-life situations.

This module uses pre-recorded audio files or text-to-speech synthesis to convert the classification result into spoken words. In systems with limited storage, a text-to-speech engine is preferred, allowing dynamic voice generation for any recognized note. The voice is clear, appropriately paced, and loud enough to be heard in both quiet and noisy environments. Users are not required to perform any actions after showing the currency to the camera, as the system seamlessly transitions to audio output. This automatic interaction improves the system’s accessibility and ensures independence for visually impaired individuals during currency identification.

To support a wide user base, the module can provide multilingual voice support, allowing the user to choose their preferred language. It also includes fallback prompts such as retry or unclear input if the system fails to recognize the note or captures an incomplete image. These prompts guide the user through the process without causing confusion or requiring technical knowledge. The module ensures that the audio is triggered only once per successful recognition, avoiding repetition or overlapping outputs. Volume and speech rate can also be adjusted according to user needs for better clarity and comfort.

The audio output module contributes directly to the usability and user-friendliness of the entire system. It ensures that the functionality remains accessible even to users unfamiliar with technology or screen-based interfaces. The design focuses on delivering accurate and timely feedback while keeping the interaction as natural as possible. This module can be used on both mobile and embedded platforms without requiring additional hardware. It plays a key role in turning the visual recognition process into an interactive experience tailored to visually impaired users. By providing clear and immediate voice responses, it completes the detection cycle and enhances user confidence and satisfaction.

5.4 MACHINE LEARNING

In the real world, we are surrounded by humans who can learn everything from their experiences with their learning capability, and we have computers or machines which work on our instructions. But can a machine also learn from experiences or past data like a human does? So here comes the role of Machine Learning. Machine Learning is said as a subset of artificial intelligence that is mainly concerned with the development of algorithms which allow a computer to learn from the data and past experiences on their own. The term machine learning was first introduced by Arthur Samuel in 1959. We can define it in a summarized way as: Machine learning enables a machine to automatically learn from data, improve performance from experiences, and predict things without being explicitly programmed.

With the help of sample historical data, which is known as training data, machine learning algorithms build a mathematical model that helps in making predictions or decisions without being explicitly programmed. Machine learning brings computer science and statistics together for creating predictive models. Machine learning constructs or uses the algorithms that learn from historical data. The more we will provide the information, the higher will be the performance.

MACHINE LEARNING ALGORITHMS:

There are various algorithms model and they are following:

1) K-Nearest Neighbours.

2) Logistic Regression.

3) Random Forest.

4) Support Vector Machine.

K-NEAREST NEIGHBORS:

K-Nearest Neighbour is one of the simplest Machine Learning algorithms based on Supervised Learning technique. K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems. K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

LOGISTIC REGRESSION:

Logistic regression is one of the most popular Machine Learning algorithms, which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables. Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.

Logistic Regression is much similar to the Linear Regression except that how they are used. Linear Regression is used for solving Regression problems, whereas Logistic regression is used for solving the classification problems. In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1). The curve from the logistic function indicates the likelihood of something such as whether the cells are cancerous or not, a mouse is obese or not based on its weight, etc. Logistic Regression is a significant machine learning algorithm because it has the ability to provide probabilities and classify new data using continuous and discrete datasets. Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification.

LR is a type of regression analysis. So, before we delve into logistic regression, let us first introduce the general concept of regression analysis.

Regression analysis is a type of predictive modeling technique which is used to find the relationship between a dependent variable (usually known as the "Y" variable) and either one independent variable (the "X" variable) or a series of independent variables. When two or more independent variables are used to predict or explain the outcome of the dependent variable, this is known as multiple regression.

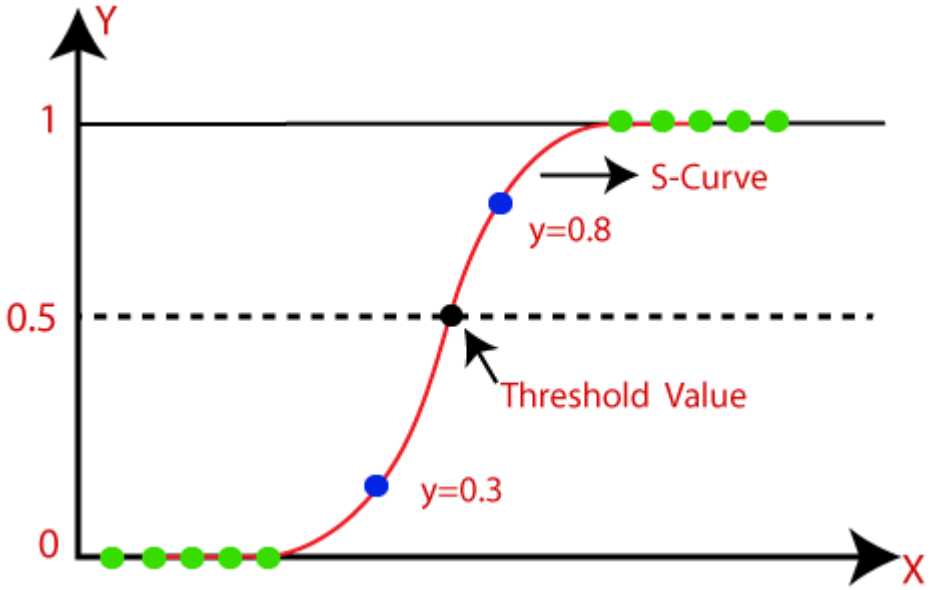


Fig no : 5.6 Logistic Regression

Forecasting the effects or impact of specific changes. For example, if a manufacturing company wants to forecast how many units of a particular product, they need to produce in order to meet the current demand.

Forecasting trends and future values. Determining the strength of different predictors, in other words, assessing how much of an impact the independent variable(s) has on a dependent variable. For example, if a soft drinks company is sponsoring a football match, they might want to

determine if the ads being displayed during the match have accounted for any increase in sales.

Regression analysis can be broadly classified into two types: Linear regression and logistic regression.

In statistics, linear regression is usually used for predictive analysis. It essentially determines the extent to which there is a linear relationship between a dependent variable and one or more independent variables. In terms of output, linear regression will give you a trend line plotted amongst a set of data points. You might use linear regression if you wanted to predict the sales of a company based on the cost spent on online advertisements, or if you wanted to see how the change in the GDP might affect the stock price of a company.

The second type of regression analysis is logistic regression, and that's what we'll be focusing on in this post. Logistic regression is essentially used to calculate (or predict) the probability of a binary event occurring.

RAMDOM FOREST:

Random Forest is a popular machine learning algorithm that belongs to the supervised learning technique. It can be used for both Classification and Regression problems in ML. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.

As the name suggests, "Random Forest is a classifier that contains a number of decision trees on various subsets of the given dataset and takes

the average to improve the predictive accuracy of that dataset." Instead of relying on one decision tree, the random forest takes the prediction from each tree and based on the majority votes of predictions, and it predicts the final output. The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.

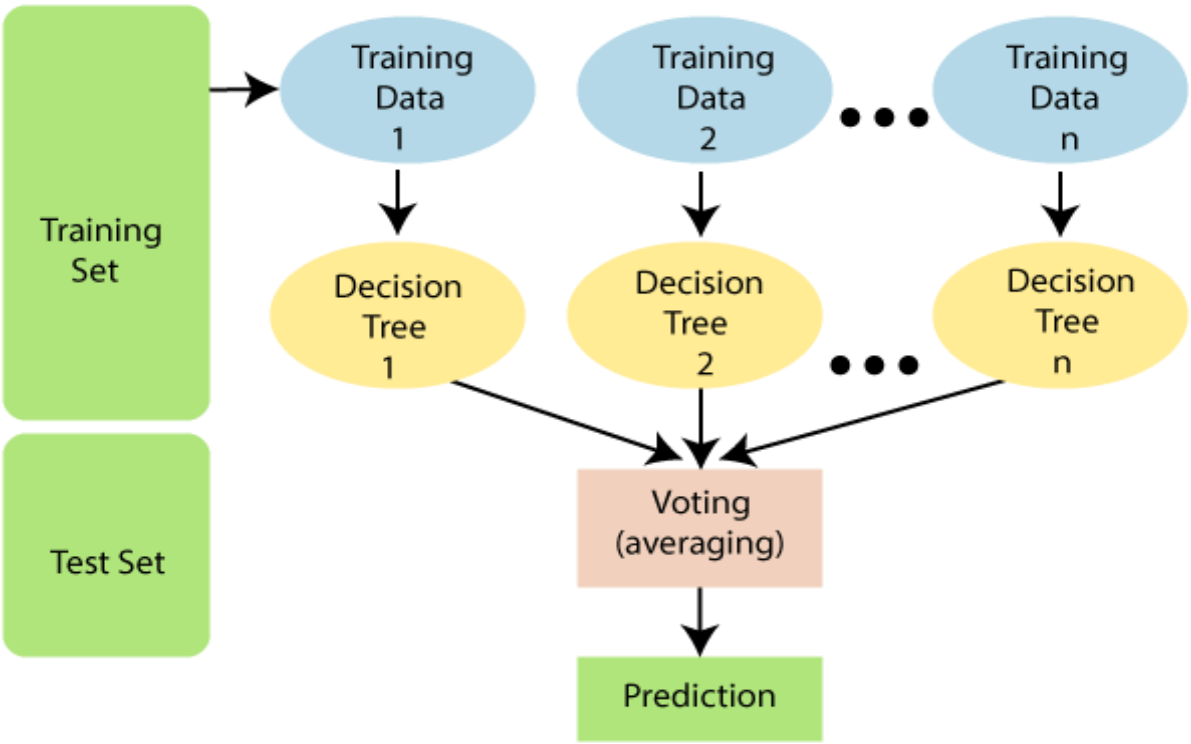


Fig.No: 5.7 WORKING OF RANDOM FOREST

SUPPORT VECTOR MACHINE

A support vector machine is a supervised learning algorithm that sorts data into two categories. It is trained with a series of data already classified into two categories, building the model as it is initially trained. The task of an SVM algorithm is to determine which category a new data point belongs in. This makes SVM a kind of non-binary linear classifier.

An SVM algorithm should not only place objects into categories, but have the margins between them on a graph as wide as possible.

It is a supervised machine learning problem where we try to find a hyperplane that best separates the two classes. Both the algorithms try to find the best hyperplane, but the main difference is logistic regression is a probabilistic approach whereas support vector machine is based on statistical approaches. There can be an infinite number of hyperplanes passing through a point and classifying the two classes perfectly. When the data is perfectly linearly separable only then we can use Linear SVM. Perfectly linearly separable means that the data points can be classified into 2 classes by using a single straight line (if 2D).

When the data is not linearly separable then we can use Non-Linear SVM, which means when the data points cannot be separated into 2 classes by using a straight line (if 2D) then we use some advanced techniques like kernel tricks to classify them. In most real-world applications we do not find linearly separable datapoints hence we use kernel trick to solve them. These are the points that are closest to the hyperplane.

A separating line will be defined with the help of these data points. It is the distance between the hyperplane and the observations closest to the

hyperplane (support vectors). In SVM large margin is considered a good margin. There are two types of margins hard margin and soft margin.

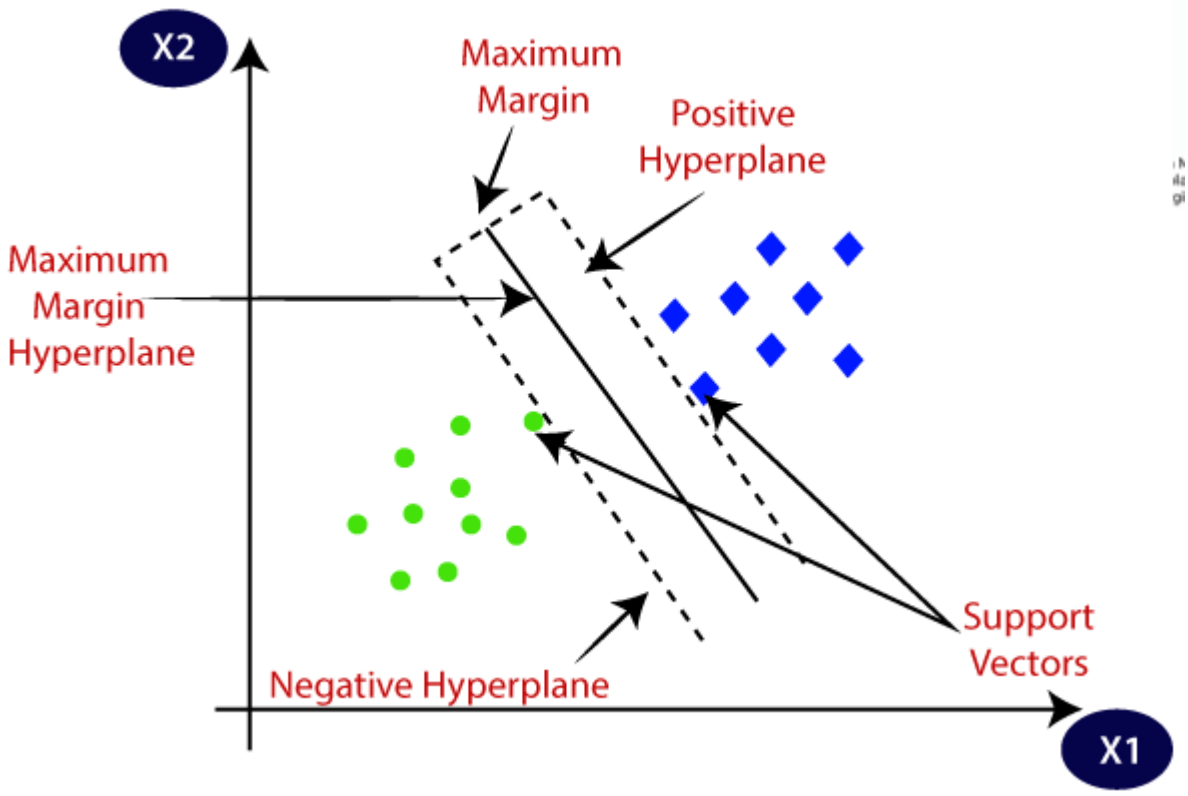


Fig.No: 5.8 SUPPORT VECTOR MACHINE

DEEP LEARNING ALGORITHMS:

Deep learning is a subset of machine learning that involves the use of artificial neural networks with multiple layers to model and learn from data. The key difference between deep learning and traditional machine learning is the use of multiple layers of non-linear processing units in deep learning algorithms, allowing them to learn complex and hierarchical representations of data.

Deep learning algorithms have been widely successful in a variety of domains, including computer vision, natural language processing, speech recognition, and robotics. They have been used to achievestate-of-the-art performance on a range of tasks such as image classification, object detection, speech recognition, machine translation, and game playing.

Some of the most popular deep learning architectures include Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Generative Adversarial Networks (GANs).

These architectures have been used to solve a range of problems, from image and speech recognition to natural language processing and autonomous driving. One of the key advantages of deep learning is that it can automatically learn relevant features and representations from raw data, without the need for manual feature engineering.

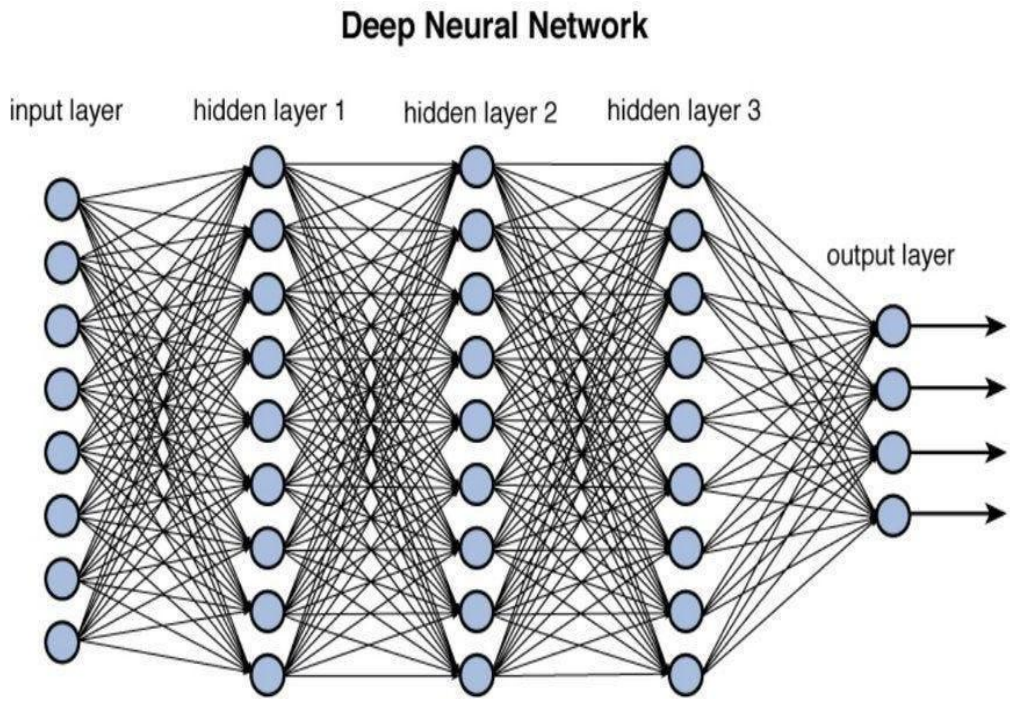


Fig.No: 5.8 NERUAL NETWORK

Convolutional neural network:

Convolutional Neural Networks (CNNs) are a type of deep learning algorithm that are particularly well-suited for image and video processing tasks, but can also be used for other types of data such as audio and text.

The key innovation of CNNs is the use of convolutional layers, which are specialized layers that learn to detect and extract local features from input images. These convolutional layers are followed by pooling layers, which aggregate and down sample the output of the convolutional layers, reducing the spatial size of the input and improving computational efficiency. The resulting feature maps are then flattened and fed into one or more fully connected layers, which perform high-level feature extraction and classification.

CNNs have achieved state-of-the-art performance on a wide range of image and video processing tasks, including image classification, object detection, semantic segmentation, and image captioning. They have also been used for speech recognition, natural language processing, and other types of data analysis tasks. Some popular CNN architectures include AlexNet, VGG, ResNet, and Inception, which have achieved groundbreaking results on various computer vision tasks.

Recurrent neural network:

Recurrent Neural Networks (RNNs) are a type of neural network that are particularly well-suited for sequence data, such as text, speech, and time series data. RNNs are designed to process sequential inputs of variable length, and they are able to capture the temporal dependencies and context in the input data.

The key innovation of RNNs is the use of recurrent connections, which allow information to be passed from one time step to the next. This means that the hidden state of the network can encode the context and history of the input sequence, which makes RNNs well-suited for tasks such as language modelling, speech recognition, and machine translation.

One limitation of basic RNNs is that they can suffer from the vanishing gradient problem, which occurs when the gradients used to update the weights during training become too small to be useful. To address this issue, various RNN variants have been proposed, including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), which use specialized gating mechanisms to control the flow of information through the network and avoid the vanishing gradient problem.

RNNs have achieved state-of-the-art performance on a wide range of sequence modelling tasks, including natural language processing, speech recognition, and time series forecasting. They are also used in combination with other types of neural networks, such as Convolutional Neural Networks (CNNs) and Transformers, to process complex inputs with multiple modalities.

TensorFlow is a powerful framework that works using a set of processing nodes, each representing a mathematical operation, the entire series of nodes called "graphs". TensorFlow is a software library for computing in graphics systems, where: The nodes in the graph represent mathematical operations. The edges in the diagram represent multidimensional data (called tensors) communicating between them.

TensorFlow is an open-source machine learning framework developed by Google. It was first released in 2015 and has since become one of the most popular deep learning frameworks in use today. TensorFlow is designed to allow developers to build and deploy machine learning models quickly and easily, using a wide range of high-level APIs and tools.

The core of TensorFlow is its computational graph engine, which allows users to define and execute complex mathematical operations on multi-dimensional arrays called tensors. The graph engine is highly optimized, and can efficiently distribute computations across multiple CPUs or GPUs for faster training and inference.

One of the key advantages of TensorFlow is its flexibility and scalability. TensorFlow can be used for a wide range of machine learning tasks, including deep learning, reinforcement learning, and probabilistic programming. It supports a variety of programming languages, including

Python, C++, and Java, and can be used on a wide range of platforms, including desktop computers, mobile devices, and cloud servers.

TensorFlow also includes a number of high-level APIs and tools that make it easier to build and deploy machine learning models. These include:

Keras: a high-level API for building and training neural networks.

TensorFlow Hub: a repository of pre-trained machine learning models and modules that can be easily integrated into new projects.

TensorFlow Lite: a lightweight version of TensorFlow designed for mobile and embedded devices.

TensorFlow.js: a JavaScript library for building and deploying machine learning models in web browsers.

TensorFlow has been used to build a wide range of machine learning applications, including image and speech recognition, natural language processing, autonomous driving, and robotics. It has also been used in research to achieve state-of-the-art results on a number of benchmark tasks.

CHAPTER 6

HARDWARE REQUIREMENT SPECIFICATION

INTEL I5 (7TH GENERATION)

The ***Intel i5 (7th generation)***, also known as ***Kaby Lake***, is a mid-range processor released by Intel in 2016. It is the successor to the 6th generation ***Skylake*** processors and was designed to offer improved performance, power efficiency, and better clock speeds. The 7th generation Intel i5 processors are built on a 14nm process technology, which allows for increased performance per watt compared to previous generations.

The ***Intel i5 (7th generation)*** CPUs typically come with 4 cores and 4 threads, although certain models, like the i5-7600K, support higher clock speeds and overclocking for more demanding tasks. These processors support ***Hyper-Threading*** in some higher-end models, which allows them to handle more threads for multitasking, although it's generally not available in the standard i5 models. The base clock speeds for these processors range from 3.0 GHz to 3.8 GHz, depending on the specific model, with the ability to turbo boost to higher frequencies for intensive tasks.

In terms of performance, the ***Intel i5 (7th generation)*** processors offer a significant performance boost over their predecessors, especially in tasks that require moderate computing power, such as gaming, productivity applications, and multitasking. These processors also support ***DDR4 memory***, providing faster memory performance than older DDR3 memory configurations. The i5 (7th generation) processors have integrated ***Intel HD Graphics 630***, which can handle light gaming and media tasks without

requiring a dedicated graphics card, making it ideal for budget-conscious users or laptops where space and power efficiency are priorities.

The ***Intel i5 (7th generation)*** is suitable for users who want a balance between performance and power efficiency. It is not as powerful as the i7 models, which come with more cores and threads, but it provides excellent performance for most everyday tasks, including gaming, content creation, and office applications. It's also a good choice for users who do not need the extreme performance of the high-end Intel i9 processors but still require a solid all-around CPU for a variety of tasks.

While the ***Intel i5 (7th generation)*** is now considered a few generations behind Intel's current processors, it still provides solid performance for many users, particularly those on a budget or who don't need the latest features. However, for users seeking more processing power or better performance in modern applications, moving to a newer generation i5 or i7 processor would provide improved performance and better energy efficiency.

8 GB RAM

8 GB RAM is a widely used memory configuration in many modern computers, offering a balanced performance for general tasks. This amount of memory is ideal for day-to-day computing activities such as web browsing, word processing, streaming media, and running multiple apps simultaneously. With **8 GB RAM**, users can open several tabs in a web browser, run office applications, and stream videos without significant lag. It is also suitable for light gaming and productivity applications, providing enough resources for a smooth user experience without overwhelming the system. For most users, 8 GB offers a great balance between performance and cost, making it one of the most common RAM configurations in mid-range computers and laptops.

While **8 GB RAM** is suitable for everyday computing, it may start to feel restrictive for more resource-intensive tasks like video editing, 3D modeling, or large-scale data analysis. Users working with multimedia content, running virtual machines, or performing computationally demanding tasks might find that 8 GB isn’t enough to deliver optimal performance. For instance, video editing software like Adobe Premiere Pro or professional photo editing tools like Photoshop can consume significant memory, leading to slowdowns if 8 GB is the only available RAM. In such cases, upgrading to 16 GB or more would provide better responsiveness and efficiency for demanding applications, ensuring smoother multitasking and faster processing.

Gaming enthusiasts also find **8 GB RAM** to be the minimum requirement for modern gaming. Many newer games, particularly those with high-quality graphics and larger environments, require more memory for

optimal performance. While **8 GB RAM** can handle most games at medium settings, it might struggle with more demanding titles or when running additional background applications. For serious gamers who want to play at higher settings or run intensive games, 16 GB or more would offer a noticeable improvement in gameplay, reducing lag and allowing smoother performance. Therefore, 8 GB is sufficient for casual gaming but may not meet the needs of users seeking top-tier gaming performance or running resource-heavy games.

For everyday tasks like web browsing, watching videos, and running office programs, **8 GB RAM** is typically more than enough. It allows users to keep multiple applications open without experiencing noticeable slowdowns, offering a fast and responsive experience for casual users. This amount of memory is also sufficient for light software development, where users may run code editors, terminals, and browsers simultaneously without performance issues. **8 GB RAM** strikes the right balance for general-purpose computing and provides excellent value for users who don't need to push the limits of their system with more demanding software.

In terms of budget and performance, **8 GB RAM** is often seen as the sweet spot for most users, particularly in laptops and mid-range desktop builds. It’s an affordable option that provides solid multitasking capabilities and ensures a smooth experience with commonly used applications. Many entry-level and mid-range laptops come with **8 GB RAM** as the standard configuration, making it an accessible option for students, professionals, and casual users. While higher-end configurations exist for those who need more power, 8 GB of RAM remains the go-to choice for users who need a reliable and efficient machine without going overboard on memory.

For laptops, **8 GB RAM** is especially useful as it provides a good balance of performance and portability. Laptops with 8 GB RAM can easily handle a variety of tasks, from office work and internet browsing to media consumption, without compromising battery life. Since laptop users often prioritize portability, **8 GB RAM** ensures that the system can perform well for most use cases while still offering long battery life. Although for heavier tasks, such as video rendering or playing graphically intense games, a higher RAM capacity may be more beneficial, 8 GB is still more than adequate for typical laptop users who need a versatile and mobile system for everyday tasks.

In summary, **8 GB RAM** is a solid and practical choice for most users, providing a balance of performance and cost. It offers enough memory to handle a wide variety of tasks, from basic computing to light gaming and content creation, without sacrificing system responsiveness. Although it may not be enough for users who engage in heavy multitasking or resource-intensive applications, for the average user, 8 GB is a sufficient and cost-effective solution. With the continual reduction in prices, 8 GB RAM remains an ideal starting point for many systems, offering good performance for users seeking an efficient and affordable computing experience.

1TB

1TB (terabyte) is a common storage size used in modern computers, laptops, and external hard drives. It represents 1,000 gigabytes (GB), providing ample space to store a significant amount of data, including documents, photos, videos, applications, and games. For typical home users, **1TB of storage** offers enough room to keep thousands of files, install several applications, and store a large library of multimedia content without worrying about running out of space. Whether for general computing or entertainment, 1TB ensures that users have plenty of space for their digital needs without constantly needing to delete old files or manage storage too frequently.

For gamers, **1TB storage** is becoming increasingly popular as modern games can be very large in size, often ranging from 40GB to over 100GB per game. With **1TB**, gamers can store several high-quality games, downloadable content, and expansion packs without worrying about storage limitations. For example, AAA titles like Call of Duty, Red Dead Redemption, or The Witcher 3, which can take up a considerable amount of space, can all be installed on a 1TB drive. This makes it a great option for gaming enthusiasts who need enough storage space to support large game libraries while also keeping performance fast and responsive.

When it comes to content creators or professionals working with multimedia, **1TB** is also a useful storage size. It allows users to store high-resolution images, 4K videos, large project files, and more without worrying about running out of storage. For photographers, videographers, and graphic designers, 1TB can be used to keep raw footage, images, design files, and backup copies of projects. Given the increasing size of high-definition

content, **1TB of storage** ensures that users can work on their projects without frequent interruptions to offload files to external drives or cloud services.

For personal use, **1TB storage** provides a balanced amount of space for daily tasks like storing documents, downloading media, and backing up personal files. It's particularly useful for users who want to have everything they need accessible on their local storage rather than relying solely on cloud services. In addition to files and media, a **1TB drive** can accommodate applications, software, and even backup files to ensure that important documents and data are safely stored locally. It's an excellent choice for users who prefer having everything in one place without worrying about running out of space too soon.

One important advantage of **1TB storage** is that it offers flexibility, especially for those who need to keep a lot of data but don't want to overpay for storage that they may not fully use. Many modern laptops, desktop computers, and external drives offer **1TB options** at an affordable price, making it an accessible and practical choice for a wide range of users. For businesses, **1TB of storage** is often the starting point for data storage, providing enough room for a team to store files, documents, and software necessary for daily operations.

In terms of **external storage**, **1TB drives** are a popular option for users looking to back up data or carry large amounts of files on the go. External drives with 1TB capacity are portable, affordable, and easy to use, making them an excellent solution for those who need extra space for their digital content. With the increase in digital media consumption, having **1TB external storage** provides an easy way to back up critical data or transfer

large files between different devices, ensuring that users don’t run into storage limitations when working with large datasets or high-definition media files.

In conclusion, **1TB of storage** provides a generous amount of space for most users, offering the ability to store a significant amount of data, including large files, games, and multimedia. Whether you’re a casual user, gamer, content creator, or business professional, **1TB** offers ample storage for everyday tasks and more intensive applications. Its flexibility, affordability, and practicality make it a common choice in a wide variety of devices, ensuring that users have enough room to store their important data without constantly managing storage capacity.

20” COLOR MONITOR

A **20-inch color monitor** is a versatile option for a variety of users, offering a good balance between screen size and space efficiency. With a 20-inch display, users can comfortably engage in everyday computing tasks like web browsing, working on documents, and watching videos. This monitor size is particularly suitable for smaller desk setups, where space might be limited. It provides enough real estate to handle multiple open windows and applications without feeling crowded, making it a great option for both home and office environments. Its compact design also allows for easy portability, meaning it can be moved between rooms or offices with ease.

For general media consumption, a **20-inch color monitor** offers a decent viewing experience. Whether you're streaming movies, watching YouTube videos, or browsing photos, the screen provides enough clarity and size for enjoyable viewing. With modern technologies such as LED

backlighting, a **20-inch monitor** can produce vibrant and bright colors, enhancing the visual experience. Additionally, this size monitor is often available with Full HD resolution (1920x1080), making it perfect for high-definition content. While not the largest display on the market, it’s still large enough to make multimedia activities engaging without overwhelming the user.

In professional environments, a **20-inch color monitor** is widely used for tasks such as word processing, spreadsheets, and web browsing. Its size strikes a perfect balance for users who need enough screen space to work efficiently but don't want a display that takes up too much room. The monitor allows users to multitask effectively by having multiple windows open simultaneously, which is important for business tasks that require reviewing several documents or monitoring different applications. The color accuracy of a **20-inch monitor** ensures that text, images, and videos are displayed with clarity and precision, making it a solid choice for office or educational work.

For light gaming, a **20-inch color monitor** provides a satisfying visual experience, although it may not be ideal for more demanding gaming setups. With its Full HD resolution, the monitor delivers crisp, clear graphics and smooth motion for less graphics-intensive games. It’s a great option for casual gamers who enjoy games with moderate system requirements. The size of the monitor also helps to provide an immersive gaming experience without overwhelming the user with an overly large display. For users who want a more compact setup while still enjoying their favorite games, the **20-inch monitor** is a balanced solution that offers good performance for gaming without the high cost of larger, more specialized gaming monitors.

The **energy efficiency** of a **20-inch color monitor** makes it a cost-effective and environmentally friendly option. These monitors typically consume less power than larger displays, which helps reduce electricity costs over time. The use of LED backlighting technology not only contributes to lower energy consumption but also provides brighter displays and longer-lasting screen life. For businesses or individuals concerned with sustainability, a **20-inch monitor** offers a green solution by minimizing both the carbon footprint and operating costs. This makes it an ideal choice for those who want to ensure that their tech usage is both efficient and environmentally conscious.

For students or those working from home, a **20-inch color monitor** can serve as an excellent tool for study, research, and video conferencing. The size provides ample space to view notes, textbooks, and research materials side by side. It’s particularly helpful for students who need to keep multiple resources open while writing essays or working on assignments. The bright, clear screen makes reading and reviewing documents comfortable for long periods without causing eye strain. For remote workers or professionals who attend virtual meetings, the monitor's clarity and color reproduction ensure a professional appearance during video calls.

In conclusion, a **20-inch color monitor** is an excellent all-around display option for a wide range of users. It provides a great balance between size and functionality, offering enough screen space for everyday tasks, media consumption, and light gaming without overwhelming the user or taking up too much desk space. The affordability and energy efficiency of these monitors make them an appealing choice for those on a budget or those looking for a reliable and compact display.

CHAPTER 7

RESULT

The developed currency note detection system successfully identifies Indian currency notes, including denominations such as ₹10, ₹20, ₹50, ₹100, ₹200 and ₹500 and provides accurate voice feedback to visually impaired users. Using OpenCV's image processing techniques, the system captures and processes the currency note, extracts key features like size, color, and security patterns, and classifies the note based on its denomination. Once identified, the system announces the denomination through voice output, ensuring ease of use in real-time across different lighting and background conditions. The solution is both accurate and user-friendly, offering visually impaired individuals the independence to manage currency effectively, with potential for integration into mobile devices or other portable platforms.

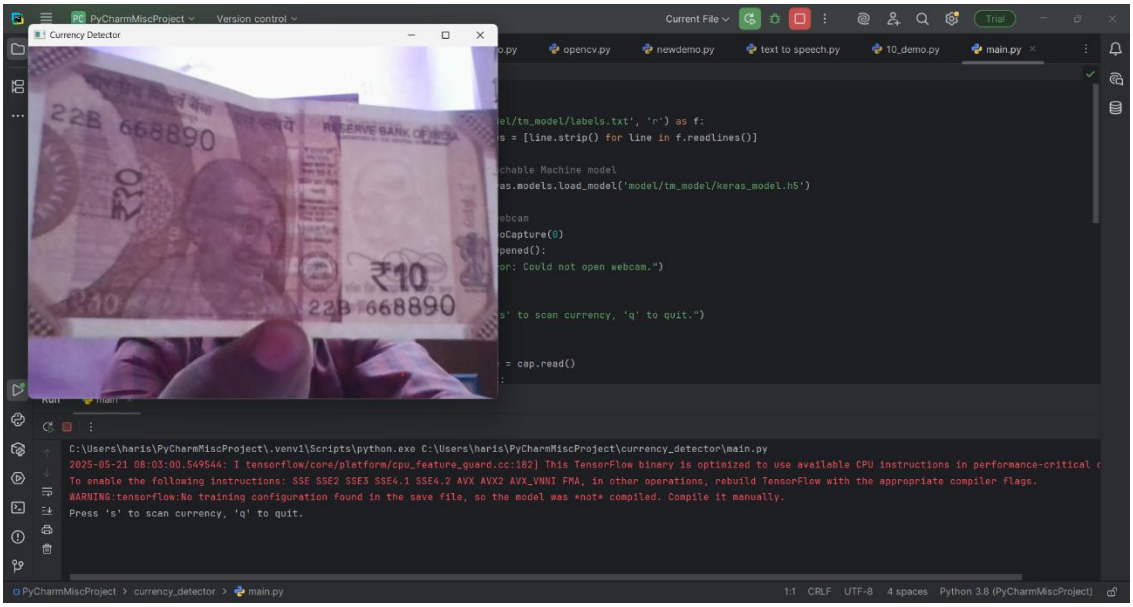
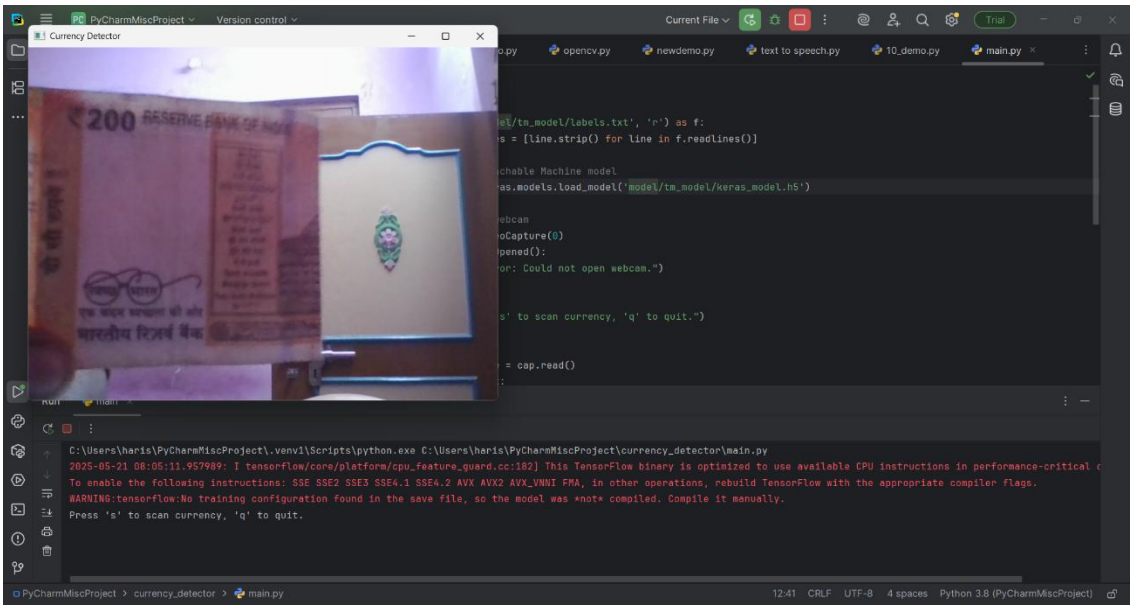


Fig No: 7.1 Prediction Output



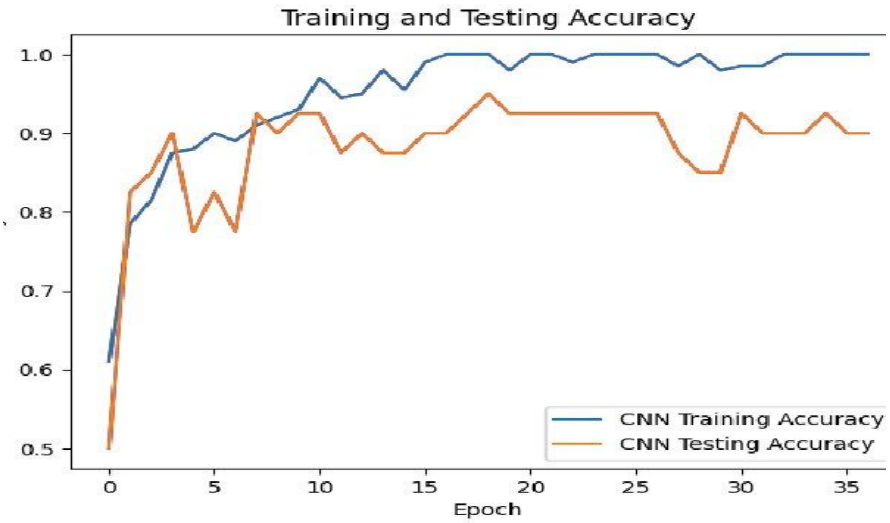


Fig No: 7.2 Training and Testing Accuracy

Fig No: 7.3 Training Phase

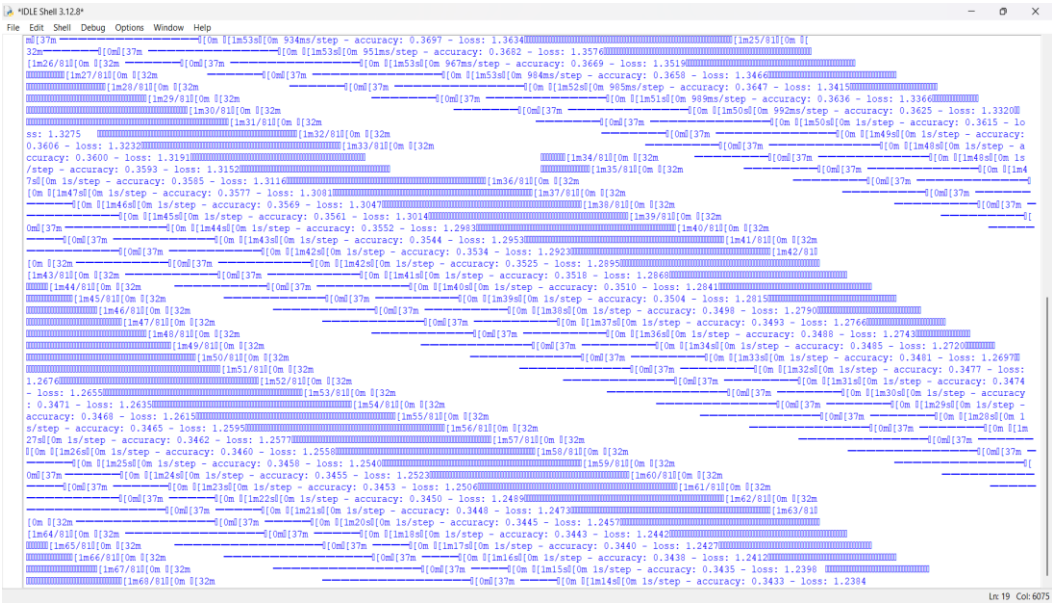


Fig No: 7.4 Predicting and Validating

CHAPTER 8

CONCLUSION

The conclusion, the currency note detection system effectively addresses the needs of visually impaired individuals by providing a reliable and accessible solution for identifying Indian currency notes. By leveraging OpenCV for image processing, the system accurately classifies different denominations and delivers real-time voice feedback, making currency management easier and more independent for users. The system performs well under varying conditions and demonstrates the potential for future integration into portable devices, further enhancing its utility. This project not only contributes to improving accessibility but also highlights the role of technology in creating inclusive solutions for visually impaired users.

CHAPTER 9

FUTURE SCOPE

The current implementation opens several avenues for future enhancement:

1. Multi-Currency Support: Extending the system to recognize foreign currency notes for use in international contexts.

2. Counterfeit Detection: Integrating counterfeit detection mechanisms by analyzing watermark and security threads more deeply.

3. Language Support: Adding multilingual audio feedback to cater to diverse users across India.

4. Cloud Integration: Utilizing cloud computing for real-time model updates and improved scalability.

5. Wearable Integration. Expanding to support smart glasses or wearable devices for seamless and hands-free recognition.

6. Offline Functionality: Enhancing offline capabilities to ensure the app works in low or no internet zones.

APPENDIX

Source Code

Keras_model.py

```
from keras.models import load_model

from PIL import Image, ImageOps

import numpy as np

# Disable scientific notation for clarity

np.set_printoptions(suppress=True)

# Load the model

model = load_model("keras_model.h5",
compile=False)

# Load the labels

class_names =open("labels.txt","r").readlines()

# Create the array of the right shape to feed into
the keras model

# The 'length' or number of images you can put
into the array is

# determined by the first position in the shape
tuple, in this case 1
```

```
data = np.ndarray(shape=(1, 224, 224, 3),
dtype=np.float32)

image = Image.open("2.jpg").convert("RGB")

# resizing the image to be at least 224x224 and
then cropping from the center

size = (224, 224)

image = ImageOps.fit(image, size,
Image.Resampling.LANCZOS)

image_array = np.asarray(image)

# Normalize the image

normalized_image_array =
(image_array.astype(np.float32) / 127.5) - 1

# Load the image into the array

data[0] = normalized_image_array

# Predicts the model

prediction = model.predict(data)

index = np.argmax(prediction)

class_name = class_names[index]

confidence_score = prediction[0][index]

# Print prediction and confidence score
```

```
print("Class:", class_name[2:], end="")

print("Confidence Score:", confidence_score)
```

Predict_img.py

```
import pyttsx3

from keras.models import load_model

import cv2

import numpy as np

import python3easypyttsx3

engine = pyttsx3.init()

engine.setProperty('rate', 150)

def speak(audio):

engine.say(audio)

engine.runAndWait()

# Disable scientific notation for clarity

np.set_printoptions(suppress=True)

# Load the model

model = load_model("keras_Model.h5",
compile=False)
```

```
# Load the labels

class_names = open("labels.txt", "r").readlines()

# CAMERA can be 0 or 1 based on default camera of
your computer

camera = cv2.VideoCapture(0)

print("Press 's' to scan currency note | Press 'q'
to quit")

while True:

# Grab the webcamera's image.

ret, image = camera.read()

keyboard_input = cv2.waitKey(1)

# Resize the raw image into (224-height,224-width)
pixels

image = cv2.resize(image, (224, 224),
interpolation=cv2.INTER_AREA)

# Show the image in a window

cv2.imshow("Webcam Image", image)

if keyboard_input == ord('s'):

    # Make the image a numpy array and reshape it
to the models input shape.
```

```
image = np.asarray(image,
dtype=np.float32).reshape(1, 224, 224, 3)

# Normalize the image array

image = (image / 127.5) - 1

# Predicts the model

prediction = model.predict(image)

index = np.argmax(prediction)

class_name = class_names[index]

confidence_score = prediction[0][index]

# Print prediction and confidence score

print("Class:", class_name[2:], end="")

print("Confidence Score:",
str(np.round(confidence_score * 100))[:-2], "%")

con_score =  str(np.round(confidence_score *
100))[:-2]

speak(f"This is a {class_name[2:]}rupees note with
{con_score}% confidence")

elif keyboard_input == ord('q'):

break

# Listen to the keyboard for presses.
```

```
#keyboard_input = cv2.waitKey(1)

# 27 is the ASCII for the esc key on your
keyboard.

# if keyboard_input == 27:

# break

camera.release()

cv2.destroyAllWindows()
```

Training_model.py

```
import os

import numpy as np

import tensorflow as tf

from tensorflow.keras.preprocessing.image import
ImageDataGenerator

from tensorflow.keras.models import Sequential

   from tensorflow.keras.layers import Conv2D,
   MaxPooling2D, Flatten, Dense, Dropout

from tensorflow.keras.optimizers import Adam

import matplotlib.pyplot as plt

# Define paths

base_dir = 'Indian currency dataset'

train_dir = os.path.join(base_dir, 'training')

val_dir = os.path.join(base_dir, 'validation')

# Image size

img_size = (224, 224)

batch_size = 16
```

```
# Data augmentation and generators

train_datagen = ImageDataGenerator(rescale=1./255,
rotation_range=10, zoom_range=0.1,
horizontal_flip=True)

val_datagen = ImageDataGenerator(rescale=1./255)

train_gen =
train_datagen.flow_from_directory(train_dir,
target_size=img_size, batch_size=batch_size,
class_mode='categorical')

val_gen = val_datagen.flow_from_directory(val_dir,
target_size=img_size, batch_size=batch_size,
class_mode='categorical')

# Class labels

class_names = list(train_gen.class_indices.keys())

print("Classes:", class_names)

# CNN model

model = Sequential([

Conv2D(32, (3, 3), activation='relu',
input_shape=(224, 224, 3)),

MaxPooling2D(2, 2),

Conv2D(64, (3, 3),
activation='relu'),MaxPooling2D(2, 2),
```

```
Conv2D(128, (3, 3), activation='relu'),

MaxPooling2D(2, 2),

Flatten(),

Dense(128, activation='relu'),

Dropout(0.5),

Dense(len(class_names), activation='softmax')

])

# Compile and train

model.compile(optimizer=Adam(),
loss='categorical_crossentropy',
metrics=['accuracy'])

history = model.fit(train_gen,
validation_data=val_gen, epochs=10)

# Save model

model.save("currency_model.h5")

# Accuracy Plot

plt.plot(history.history['accuracy'], label='Train
Acc')

plt.plot(history.history['val_accuracy'],
label='Val Acc')
```

```
plt.title("Accuracy")

plt.legend()

plt.show()
```

CNN_Model.py

```
import os

import numpy as np

from tensorflow.keras.preprocessing.image import
ImageDataGenerator

from tensorflow.keras.models import Sequential

from tensorflow.keras.layers import Conv2D,
MaxPooling2D, Flatten, Dense

from tensorflow.keras.optimizers import Adam

from tensorflow.keras.callbacks import
ModelCheckpoint

# Paths

dataset_path = "D:\project code
currency\currency_recognition_system
(2)\currency_recognition_system\dataset"

model_path = "models/cnn_model.h5"

# Image parameters
```

```
img_width, img_height = 100, 100

batch_size = 16

epochs = 10

# Data preparation

datagen = ImageDataGenerator(rescale=1./255,
validation_split=0.2)

train_gen = datagen.flow_from_directory(

dataset_path,

target_size=(img_width, img_height),

batch_size=batch_size,

class_mode='categorical',

subset='training')

val_gen = datagen.flow_from_directory(

dataset_path,

target_size=(img_width, img_height),

batch_size=batch_size,

class_mode='categorical',

subset='validation'

)
```

```
# Build CNN model

model = Sequential([

Conv2D(32, (3, 3), activation='relu',
input_shape=(img_width, img_height, 3)),

MaxPooling2D(2, 2),

Conv2D(64, (3, 3), activation='relu'),

MaxPooling2D(2, 2),

Flatten(),

Dense(64, activation='relu'),

Dense(2, activation='softmax')

])

model.compile(optimizer=Adam(),
loss='categorical_crossentropy',
metrics=['accuracy'])

model.fit(train_gen,

validation_data=val_gen,

epochs=epochs)

model.save(model_path)

print(f"Model saved to {model_path}")
```

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